Fixed Dynamics Disprove Two Cesàro-Residuality Questions

Run `fa_banach_001`, lane 3

22 July 2026

Status. Candidate counterexamples, likely valid, to Questions 5.6 and 5.9 as stated in Badea–Grivaux [1]. These are literal-formulation counterexamples: they do not answer the narrower and substantive case in which the acting dynamics are required to be independent of the invariance dynamics.

1 Source questions

For an integer $m \geq 1$, let T_m be the pushforward on probability measures induced by $x \mapsto mx \pmod 1$ on $\mathbb{T}$. Let $\mathcal{P}_p(\mathbb{T})$ be the compact space of T_p-invariant probability measures, and let $\mathcal{P}_{p,c}(\mathbb{T})$ denote its nonatomic members. Section 5.b of [1] defines

$$G''_{p,(c_n)} = \left\{ \mu \in \mathcal{P}_{p,c}(\mathbb{T}) : \frac{1}{N} \sum_{n=0}^{N-1} T_{c_n}\mu \xrightarrow{w^*} \mathrm{Leb} \right\}.$$

Question 5.6 asks whether this set is residual in $\mathcal{P}_p(\mathbb{T})$ for every $p \geq 2$ and every strictly increasing integer sequence $(c_n)_{n \geq 0}$.

In dimension $d \geq 2$, for nonsingular integer matrices A, B, the paper similarly defines

$$G''_{A,B} = \left\{ \mu \in \mathcal{P}_{A,c}(\mathbb{T}^d) : \frac{1}{N} \sum_{n=0}^{N-1} T_{B^n}\mu \xrightarrow{w^*} \mathrm{Leb}_d \right\}.$$

Question 5.9 asks whether $G''_{A,B}$ is residual in $\mathcal{P}_A(\mathbb{T}^d)$ for every such pair A, B. The exact statements occur on article pages 28–29:

Question 5.5. Let $p \geq 2$, and let $(c_n)_{n\geq 0}$ be any strictly increasing sequence of integers. Is it true that the set $G_{p,(c_n)}$ is residual in $(\mathcal{P}_p(\mathbb{T}), w^*)$?

If μ is (c_n)-generic, then $\frac{1}{N}\sum_{n=0}^{N-1} T_{c_n}\mu \xrightarrow{w^*} \mathrm{Leb} \quad \text{as } N \to +\infty$. It is thus natural to consider the set

$$G''_{p,(c_n)} = \left\{\mu \in \mathcal{P}_{p,c}(\mathbb{T})\ ;\ \frac{1}{N}\sum_{n=0}^{N-1} T_{c_n}\mu \xrightarrow{w^*} \mathrm{Leb} \quad \text{as } N \longrightarrow +\infty\right\}$$

and to ask the following question.

Question 5.6. Let $p \geq 2$, and let $(c_n)_{n\geq 0}$ be a strictly increasing sequence of integers. Is the set $G''_{p,(c_n)}$ residual in $(\mathcal{P}_p(\mathbb{T}), w^*)$?

For the sequences considered in [24, 27, 35, 38], the density of the set $G''_{p,(c_n)}$ in $(\mathcal{P}_p(\mathbb{T}), w^*)$ follows from the result that ergodic measures of positive entropy in $\mathcal{P}_p(\mathbb{T})$ are (c_n)-generic. But the question of the residuality remains widely open, and it is actually not known if the set $G''_{p,(q^n)}$ is residual in $(\mathcal{P}_p(\mathbb{T}), w^*)$ when $q \geq 2$ is an integer which is multiplicatively independent from p. This question is also connected to another conjecture from [37], called (C7), which runs as follows and seems to be still open:

Conjecture 5.7. Let $p, q \geq 2$ be multiplicatively independent integers. For any measure $\mu \in \mathcal{P}_{p,c}(\mathbb{T})$, μ-almost every $x \in \mathbb{T}$ is normal in base q, i.e.

$$\frac{1}{N}\sum_{n=0}^{N-1} e^{2i\pi a q^n x} \to 0 \quad \text{as} \quad N \to \infty \quad \text{for every } a \in \mathbb{Z} \setminus \{0\}.$$

In our examination of Conjecture 1.4 in the multidimensional context, we have established conditions on matrices $A, B \in M_d(\mathbb{Z})$ with non-zero determinant that imply that the set

$$G_{A,B} = \left\{\mu \in \mathcal{P}_{A,c}(\mathbb{T}) \; ; \; T_{B^n}\mu \overset{w^*}{\not\to} \mathrm{Leb}_d\right\}$$

is residual in $(\mathcal{P}_A(\mathbb{T}^d), w^*)$. These conditions (that A be diagonalisable in $M_d(\mathbb{C})$, that $\det A$ and $\det B$ be relatively prime...) arise due to technical difficulties in the proofs in the higher dimensional case. However, it may be that these conditions are not necessary; this is true in the one-dimensional setting.

Question 5.8. Let $d \geq 2$ and $A, B \in M_d(\mathbb{Z})$ with $\det A \neq 0$ and $\det B \neq 0$. Is it true that the set $G_{A,B}$ is residual in $(\mathcal{P}_A(\mathbb{T}^d), w^*)$?

Let also

$$G''_{A,B} = \left\{\mu \in \mathcal{P}_{A,c}(\mathbb{T}^d) \; ; \; \frac{1}{N}\sum_{n=0}^{N-1} T_{B^n}\mu \xrightarrow{w^*} \mathrm{Leb}_d \quad \text{as } N \longrightarrow +\infty\right\}.$$

In analogy to Question 5.6 , one may also ask:

Question 5.9. Let $d \geq 2$ and $A, B \in M_d(\mathbb{Z})$ with $\det A \neq 0$ and $\det B \neq 0$. Is it true that the set $G''_{A,B}$ is residual in $(\mathcal{P}_A(\mathbb{T}^d), w^*)$?

References

2 Proof Intuition

Both questions quantify over dynamics that are allowed to coincide with the dynamics defining invariance. If μ is T_p-invariant and $c_n = p^n$, then every term $T_{c_n}\mu$ equals μ. The Cesàro averages therefore cannot wash μ out to Lebesgue measure: they are the constant sequence μ. Hence the proposed generic set is only the Lebesgue measure itself. The same fixed-point obstruction works in higher dimension by choosing $B = A$; the concrete choice $A = B = 2I_d$ already suffices.

3 Counterexamples

Theorem 1. *The answers to Questions 5.6 and 5.9 of [1], as stated, are both negative.*

More precisely:

1. *For every $p \geq 2$, if $c_n = p^n$, then $G''_{p,(c_n)} = \{\mathrm{Leb}\}$, which is not residual in $\mathcal{P}_p(\mathbb{T})$.*
2. *For every $d \geq 2$, if $A = B = 2I_d$, then $G''_{A,B} = \{\mathrm{Leb}_d\}$, which is not residual in $\mathcal{P}_A(\mathbb{T}^d)$.*

Proof. Pushforwards under multiplication obey $T_a T_b = T_{ab}$. Fix $p \geq 2$ and put $c_n = p^n$. This is a strictly increasing sequence of integers. If $\mu \in \mathcal{P}_p(\mathbb{T})$, then $T_p\mu = \mu$, and induction gives

$$T_{c_n}\mu = T_{p^n}\mu = T_p^n\mu = \mu \qquad (n \geq 0).$$

Consequently, for every $N \geq 1$,

$$\frac{1}{N}\sum_{n=0}^{N-1} T_{c_n}\mu = \mu.$$

For a nonatomic μ, these averages converge weak-star to Leb if and only if $\mu = \mathrm{Leb}$. Thus $G''_{p,(p^n)} = \{\mathrm{Leb}\}$.

The weak-star space $\mathcal{P}_p(\mathbb{T})$ is Hausdorff and contains the distinct point mass δ_0, since $T_p\delta_0 = \delta_0$. Therefore the singleton $\{\mathrm{Leb}\}$ is not dense. A residual subset, in the convention of the source paper, must contain a dense G_δ set, so $G''_{p,(p^n)}$ is not residual. This disproves Question 5.6.

Now fix $d \geq 2$ and take $A = B = 2I_d$. Both matrices are integral and have determinant $2^d \neq 0$, so they meet all hypotheses of Question 5.9. For every $\mu \in \mathcal{P}_A(\mathbb{T}^d)$,

$$T_{B^n}\mu = T_{A^n}\mu = T_A^n\mu = \mu.$$

The same averaging identity shows that $G''_{A,B} = \{\mathrm{Leb}_d\}$. But $\mathcal{P}_A(\mathbb{T}^d)$ also contains the distinct invariant point mass δ_0, so this singleton is not dense and hence is not residual. This disproves Question 5.9. $\square$

4 Verification and scope

Every hypothesis is visible in the construction. The sequence $(p^n)_{n\geq 0}$ is strictly increasing; $2I_d$ is a nonsingular integer matrix; Leb and Leb_d are nonatomic and invariant; and δ_0 provides an explicit second point in each ambient invariant-measure space. No external theorem beyond the definitions is used.

The paper immediately singles out the case $c_n = q^n$ with q multiplicatively independent from p as widely open. Our example has $q = p$ and therefore does not settle that intended independence regime. Likewise, it does not address a multidimensional variant imposing a genuine independence or coprimality condition between A and B. Questions 5.5 and 5.8, which concern failure of pointwise weak-star convergence rather than convergence of Cesàro averages, are not contradicted.

5 Novelty check

The run indexes were searched for arXiv:2303.01089, the paper title, Questions 5.6 and 5.9, Cesàro residuality, and the fixed sequences $c_n = p^n$ and $B = A$. No prior packet or attempt was found. Bounded web searches on 22 July 2026 used the exact question numbers with the arXiv id, the authors' names with the displayed G'' notation, and the phrases `c_n = p^n`, `T_p-invariant`, and `Cesaro residual`. They found the source paper and its journal page but no later answer or discussion of this obstruction. Novelty confidence is moderate: the proof is elementary and may be an unrecorded formulation correction.

6 Human-review recommendation

Review as a likely valid literal counterexample to both universal questions. The decisive checks are that the published Questions 5.6 and 5.9 impose no independence condition, and that residuality is taken in the full spaces $\mathcal{P}_p(\mathbb{T})$ and $\mathcal{P}_A(\mathbb{T}^d)$, respectively.

References

[1] C. Badea and S. Grivaux, *Around Furstenberg's times p, times q conjecture: times p-invariant measures with some large Fourier coefficients*, Discrete Analysis 2024:10 (2024), 31 pp., doi:10.19086/da.124611; arXiv:2303.01089.